E-Commerce Sales Analysis

Project Report | Tableau Dashboard

An end-to-end overview of the dataset, business analysis areas, dashboard design, and reproducible project workflow.

Dataset profile	Value
Rows	115,746
Columns	20
Date coverage	01 Jan 2016 – 31 Dec 2017
Missing values (all columns)	1,027
Duplicate rows	0

Prepared from the provided Ecommerce Sales.csv dataset and the Tableau workbook structure. Findings are described conservatively where the workbook exposes chart design but not a rendered dashboard image.

1. Overview

This project explores e-commerce sales and order data to review commercial performance across product categories, markets, order quantities, and customer acquisition trends. The accompanying Tableau workbook organizes the analysis into KPI views and comparative visualizations to make business metrics easier to review.

Project objective

Turn transaction-level data into clear visual summaries that help stakeholders monitor performance, compare markets and categories, and identify areas for further investigation.

2. Dataset

The supplied CSV contains **115,746 rows** and **20 columns**. Its fields include order dates and identifiers, category, customer segment and location, market and region, order quantity, product price, sales, profit per order, and profit margin.

Metric (source column)	Value
Total of Sales column	27,542,886.86
Total Profit Per Order	2,614,204.18
Total Order Quantity	295,330

Data quality checks

An initial check found 1,027 missing cell(s) and 0 fully duplicated row(s). These are profiling results; whether to remove or impute records should be decided based on business context before final publication.

3. Tools

Tableau for dashboarding and visual analysis; **CSV** as the source data format. The workbook includes separate worksheets for category comparison, market share, customer acquisition by month, and KPI views for sales, profit, and order quantity.

4. Analysis Steps

Load data: Connect the Tableau workbook to Ecommerce Sales.csv and verify that fields are mapped correctly.

Profile and prepare: Review column types, missing values, duplicate rows, and date parsing; apply appropriate cleaning only where justified.

Explore performance: Compare sales and profit measures across categories and markets, and review order quantity.

Review customer trends: Examine customer acquisition patterns by market and month.

Build dashboard: Combine KPI sheets and supporting charts into a single dashboard for quick review.

Document findings: Use dashboard observations to prepare a concise report and business recommendations, validating all conclusions against the underlying data.

5. Tableau Dashboard

The provided workbook contains a dashboard named “Dashboard 1” and the following worksheets:

Worksheet	Purpose
Sales KPI	High-level sales metric

Worksheet	Purpose
Profit KPI	High-level profit metric
Order Quantity KPI	Order quantity overview
Category-wise YTD vs PYTD Sales	Compare year-to-date and prior-year-to-date sales by category
Market Share by Sales	Review sales contribution across markets
Market-wise Customer Acquisition per Month	Explore customer acquisition trends over time and by market

Dataset Snapshot: Category Summary

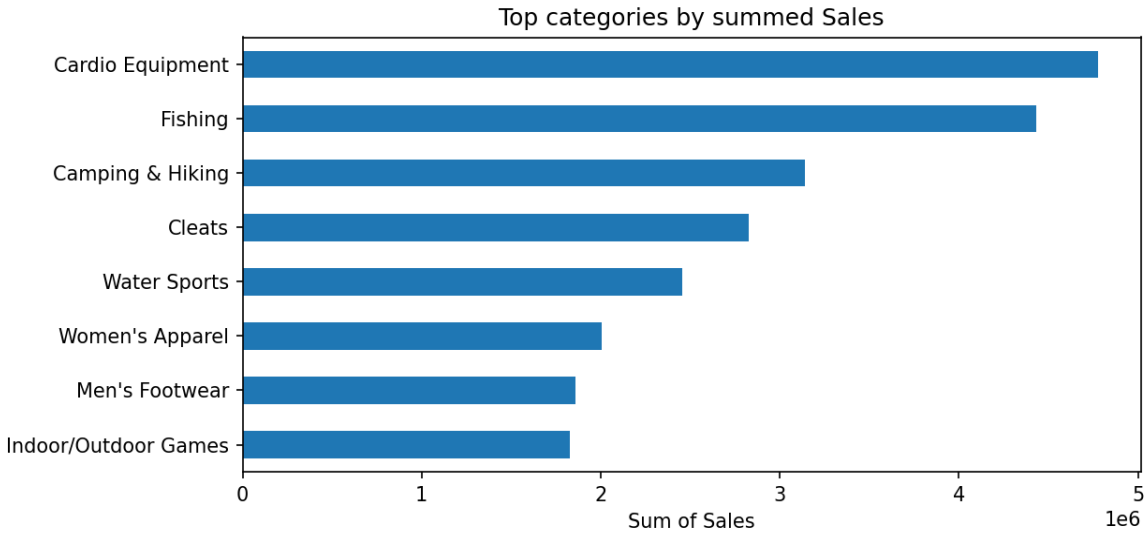


Chart groups the top categories by summed Sales, based on the source CSV. Confirm the intended business definition of the Sales field before interpreting this as revenue.

6. Results and Business Value

The workbook is structured to support these review areas:

• Compare category-level sales performance across current and prior periods.
• Review how sales are distributed across markets.
• Track monthly customer acquisition patterns.
• Monitor sales, profit, and order quantity through KPI views.

Recommended next step: use the dashboard to identify specific categories or markets that warrant attention, then validate the underlying values and document evidence-based actions. This report does not assert a specific winning market or category without a verified, rendered dashboard result.

7. How to Run

1. Keep *Ecommerce Sales.csv* and *Ecommerce Sales Dashboard.twb* in an accessible folder.
2. Open the TWB workbook in Tableau Desktop.
3. If Tableau reports a missing data source, edit the connection and select the provided CSV file.
4. Refresh the data and check date and numeric field types.
5. Open “Dashboard 1” and interact with the available visualizations.
6. Save a copy of the workbook after confirming that all sheets render correctly.

8. Project Files

• Ecommerce Sales.csv — source dataset
• Ecommerce Sales Dashboard.twb — Tableau workbook
• Ecommerce_Sales_Report.pdf — project documentation

Skills Demonstrated

Data profiling • Data preparation • Tableau • KPI design • Data visualization • Sales analysis • Market analysis • Business reporting

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